Ctrl + S

```
using Plots


```

```
g (generic function with 1 method)
```

```
begin
a = 0.967
M = 0.01
F(E) = E - a * sin(E) - M
dF(E) = 1 - a * cos(E)
g(E) = M + a * sin(E)
end
```

```
# Punto fijo
let
E = 0.01
tolerancia =  1e-8
n=0
while abs(F(E)) > tolerancia && n < 1000
    E = g(E)
    n = n+1
end
println("Punto fijo: ", E,"pasos: ", n)
end
```

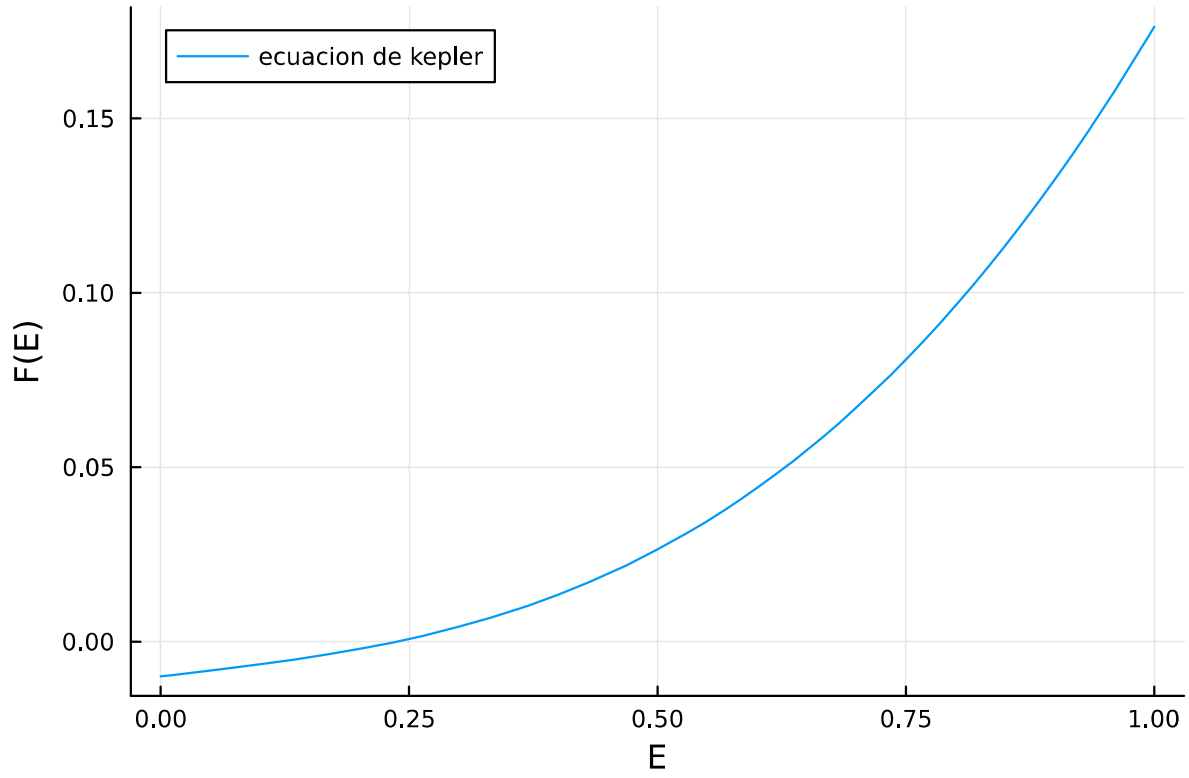
```
Punto fijo: 0.23765798700248433pasos: 236
```

```
# Newton Raphson
let
    E = 0.5
    tolerancia = 1e-8
    n = 0
while abs(F(E)) > tolerancia && n < 1000
    E = E - F(E)/dF(E)
    n = n + 1
end
    println("Newton rapshon: ", E, "pasos: ", n)
end
```

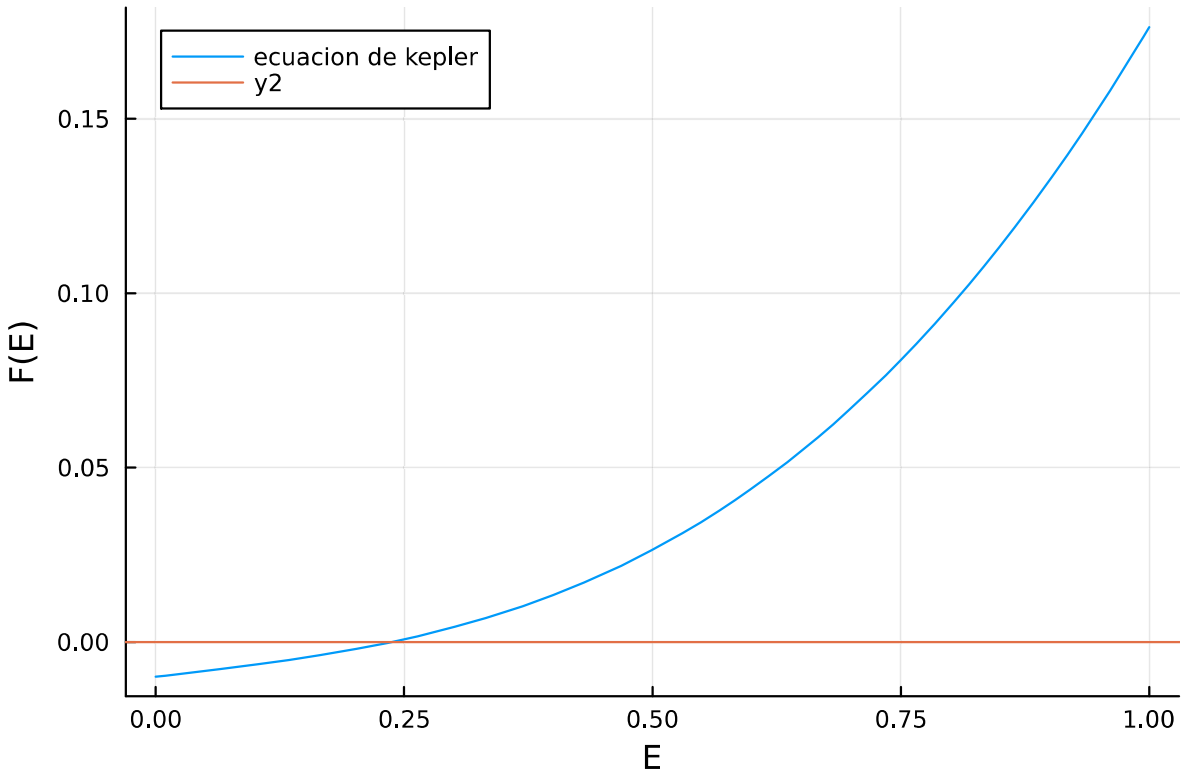
```
Newton rapshon: 0.23765814412142724pasos: 5
```

```julia
#Secante
let
    E0 = 0.01
    E1 = 0.1
    tolerancia = 1e-8
    n = 0
    while abs(F(E1)) > tolerancia && n < 1000
        E2 = E1 - F(E1)*(E1-E0)/(F(E1)-F(E0))
        E0 = E1
        E1 = E2
        n=n+1
    end
    println("Secante: ", E1, "pasos: ", n)
end
```

```
Secante: 0.23765807267978026pasos: 5
```




```julia
plot(F, 0, 1, label="ecuacion de kepler", xlabel="E", ylabel="F(E)")
```




```
1 hline!([0])
```

```
1 plot()
```